

(S)-联萘酚钛体系催化 3-噻吩铝试剂对酮的不对称加成反应

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摘要 含噻吩基团的手性芳香醇是自然界重要的活性物质,其基本结构单元广泛地存在于生物活性物质和医药中。报道了(S)-联萘酚和 $\text{Ti}(\text{O}^i\text{Pr})_4$ 催化 3-噻吩基铝试剂对酮的不对称加成反应,以较高的产率和优异的立体选择性(高达 91% *ee*)得到相应的含有 3-噻吩基乙醇。该催化体系具有广泛的适应性,能够适合含有吸电子和供电子基团的各种酮的不对称加成,得到相应的 3-噻吩基乙醇。

关键词 不对称合成; 噻吩基; 铝试剂; 钛; 酮

Asymmetric Addition of 3-Thienyl Aluminum to Ketones Catalyzed by the Simple Titanium Catalytic System of (S)-1,1'-Binaphthol

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Abstract The optically active diaryl alcohols bearing thienyl group are well-known for their biological activity as well as key substructure in bioactive compounds and pharmaceuticals. The asymmetric addition of 3-thienyl aluminum to ketones afforded the optically active 3-thienyl arylethanol in good yields with high enantioselectivities of up to 91% *ee* employing the simple titanium catalytic system of (S)-1,1'-binaphthol. The catalytic system could also suit for various ketones bearing an electron-donating or electron-withdrawing groups to afford the corresponding diarylethanol containing substituted 3-thienyl groups.

Keywords asymmetric synthesis; thienyl; aluminum; titanium; ketones

1 Introduction

The enantioselective addition of carbon-based nucleophiles to organic carbonyl compounds provides the most straightforward strategy for the construction of optically active alcohols.^[1~5] The optically active diaryl alcohols bearing heteroaryl such as furyl, thienyl, pyridyl or indolyl groups are well-known for their biological activity as well as key substructure in bioactive compounds and pharmaceuticals.^[6] However, their syntheses were sporadically reported in a few papers via addition of organometallic reagents to heteroaryl-substituted carbonyl compounds or reduction of diaryl ketones bearing heteroaryl groups.^[7] The catalytic enantioselective addition of heteroaryl nucleophiles to organic carbonyl compounds provided a systematic approach to synthesize chiral heteroaryl alcohols.

The optically active furyl alcohols have been realized

through the asymmetric addition of furyl aluminum reagent^[8] or furyltitanium reagent^[9] to ketones in the presence of 10 mol% (S)-1,1'-binaphthol ((S)-BINOL). The optically active thienyl alcohols were also reported through the asymmetric additions of thienylboronic acid^[10] or thienyl-aluminum reagents^[11] to organic carbonyl compounds in good yields and high enantioselectivities. Recently, heteroaryl zincs^[12] and titanium reagents^[13] bearing thienyl, benzothienyl, furyl and indolyl groups were also applied to asymmetric addition of aldehydes affording aryl heteroaryl- or diheteroaryl-methanol derivatives in high enantioselectivities. More recently, we have also reported the catalytic asymmetric addition of pyridylaluminums to aldehydes catalyzed by a titanium catalytic system of (R)-H₈-BINOLate.^[14] As a continuous research on asymmetric heteroaryl addition to organic carbonyl compounds, we herein report the catalytic asymmetric addition of 3-thienyl

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aluminum to ketones catalyzed by a titanium catalytic system of (*S*)-BINOL to afford the chiral diarylmethanols bearing 3-thienyl group.

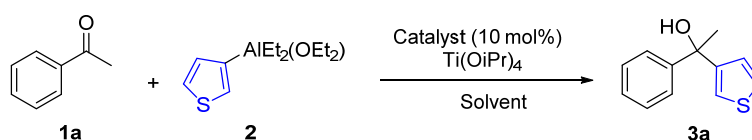
2 Results and discussion

Asymmetric addition reactions of (3-thienyl)AlEt₂(OEt₂) to acetophenone (**1a**) were first optimized, and the results are summarized in Table 1. Tuning the amounts of Ti(O^{*i*}Pr)₄ and (3-thienyl)AlEt₂(OEt₂), the optimal reaction conditions of 3.5 equiv. of Ti(O^{*i*}Pr)₄ and 3.0 equiv. of (3-thienyl)AlEt₂(OEt₂) in the presence of 10 mol% (*S*)-BINOL in toluene at 0 °C were found to afford product **3a** in a 88% yield with an excellent 90% *ee* (Entry 7). The investigations for the choice of the solvent showed that toluene was the best one (Entries 7, 9~12). For the optimal catalytic systems, different chiral ligands (Figure 1) were

test for asymmetric addition reactions (Entries 7, 13~16). Results showed that the commercially available and relatively cheap chiral ligand, (*S*)-BINOL, was the most efficient one to afford the chiral alcohol **3a** in a 88% yield with an excellent 90% *ee* (Entry 7).

With the optimized conditions established, the asymmetric addition of 3-thienyl aluminum reagent to functionalized ketones were then studied, and the results were listed in Table 2. For aromatic ketones with either an electron-withdrawing substituent at the aryl ring, 3-thienyl additions afforded tertiary alcohols in high yields with excellent enantioselectivities of 90% *ee* or higher (Entries 1~7), except for substrates of 2'-nitroacetophenone (**1b**) and 4'-bromoacetophenone (**1e**), which afforded **3b** and **3e** in 88% and 86% *ee* (Entries 2 and 5). However, for aromatic ketones with an electron-donating substituent at the aryl ring,

Table 1 Optimization of enantioselective addition of 3-thienyl aluminum to acetophenone^a



Entry	Ligand	Ti(O ^{<i>i</i>} Pr) ₄ ^b /equiv.	(3-Thienyl)Al ^b /equiv.	Solvent	Yield/%	<i>ee</i> ^c /%
1	(<i>S</i>)-BINOL	5.0	1.5	Toluene	21	82
2	(<i>S</i>)-BINOL	5.0	2.0	Toluene	31	85
3	(<i>S</i>)-BINOL	5.0	2.5	Toluene	53	84
4	(<i>S</i>)-BINOL	5.0	3.0	Toluene	35	89
5	(<i>S</i>)-BINOL	5.0	3.5	Toluene	40	82
6	(<i>S</i>)-BINOL	2.5	3.0	Toluene	70	62
7	(<i>S</i>)-BINOL	3.5	3.0	Toluene	88	90
8	(<i>S</i>)-BINOL	4.0	3.0	Toluene	50	88
9	(<i>S</i>)-BINOL	3.5	3.0	THF	62	84
10	(<i>S</i>)-BINOL	3.5	3.0	<i>n</i> -Hexane	63	88
11	(<i>S</i>)-BINOL	3.5	3.0	Et ₂ O	93	81
12	(<i>S</i>)-BINOL	3.5	3.0	CH ₂ Cl ₂	85	31
13	A	3.5	3.0	Toluene	50	25
14	B	3.5	3.0	Toluene	40	-10
15	C	3.5	3.0	Toluene	60	-51
16	D	3.5	3.0	Toluene	45	15

^a Condition: acetophenone (1.00 mmol), solvent (5.0 mL), time: 12 h. ^b Equivalents of Ti(O^{*i*}Pr)₄ and **2** are relative to acetophenone. ^c Enantioselectivities were determined by HPLC.

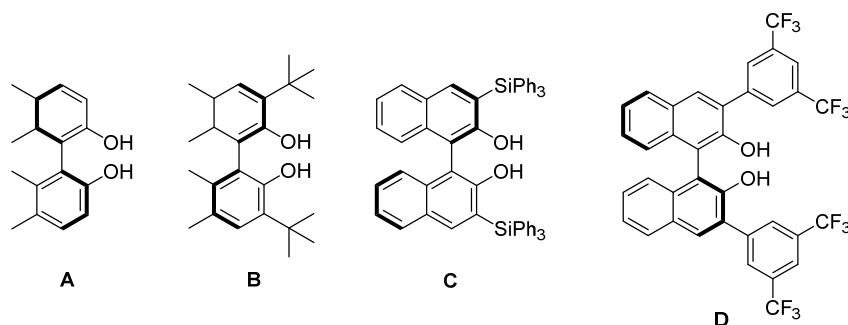
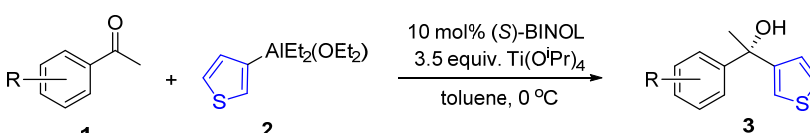
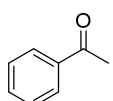
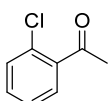
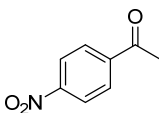
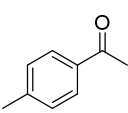
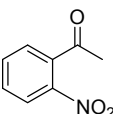
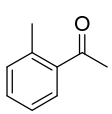
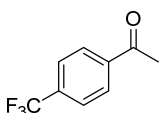
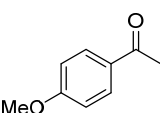
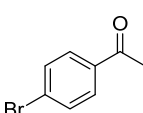
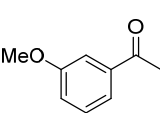
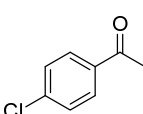
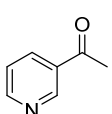


Figure 1 Different chiral ligands

Table 2 Enantioselective addition of 3-thienyl aluminum reagent to ketones^a

									
Entry	Ketone	Product	Yield/%	ee/%	Entry	Ketone	Product	Yield/%	ee/%
1		3a	88	90	7		3g	80	90
2		3b	91	91	8		3h	80	85
3		3c	90	88	9		3i	78	85
4		3d	90	90	10		3j	83	90
5		3e	82	86	11		3k	85	70
6		3f	87	90	12		3l	79	91

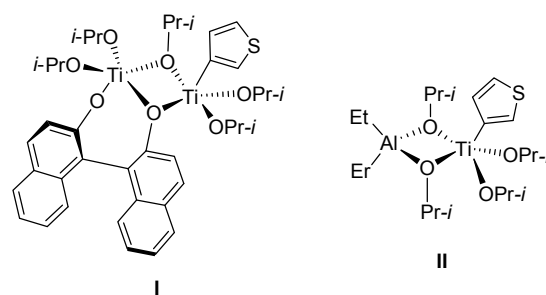
^a Condition: (S)-BINOL (0.10 mmol), Ti(OⁱPr)₄ (3.50 mmol), **2** (3.00 mmol), ketone (1.00 mmol), toluene (5.0 mL). ^b Enantioselectivities were determined by HPLC.

3-thienyl additions afforded tertiary alcohols in high yields with slight lower enantioselectivities of 88% ~ 70% *ee* (Entries 8 ~ 11). For heteroaromatic ketones, 3-thienyl addition to 1-(pyridin-3-yl)ethanone furnished the product **3l** in a good yield with a excellent enantioselectivity of 90% *ee* (Entry 12).

Based on the previous study of titanium-catalyzed asymmetric arylation of organozinc, organoaluminum, or organotitanium, the catalytic reactions involve the similar active species which are proposed to be dititanium species **I** containing one BINOLate ligand and the nucleophile.^[4a,15] The roles of Ti(OⁱPr)₄ in the reaction not only gave the above active dititanium species, but also reacted with 3-thienyl aluminum reagent to afford an immediate of a bimetallic complex **II**.^[4a]

3 Conclusions

In summary, the catalytic enantioselective heteroarylation of 3-thienyl aluminum nucleophiles with various ketones has been successfully developed employing the simple titanium catalytical system of (S)-BINOL to afford a series of chiral arythanols containing 3-thienyl groups in high yields with excellent enantioselectivities of up to 91% *ee*.



4 Experimental section

4.1 General methods

All syntheses and manipulations of air- and moisture-sensitive materials were performed under a dry argon atmosphere using standard Schlenk techniques or in a glovebox. Solvents were refluxed and distilled over sodium/benzophenone under argon prior to use. ¹H NMR and ¹³C NMR spectra in CDCl₃ were recorded on a Bruker AV-300 NMR spectrometer with chemical shifts from the internal TMS. HPLC analyses were recorded with an Agilent 1200 Series spectrometer using Chiralpak column.

4.2 General procedure for the asymmetric addition of 3-thienyl aluminum to ketones

Under a dry nitrogen atmosphere, (*S*)-BINOL (28.6 mg, 0.1 mmol) and $\text{Ti}(\text{O}^i\text{Pr})_4$ (1.04 mL, 3.5 mmol) were mixed in dry toluene (5.0 mL) at room temperature. After stirring for 30 min, (3-thienyl) $\text{AlEt}_2(\text{OEt}_2)$ (3.0 mmol) was added to the resulting solution at 0 °C. The mixture was stirred for another 10 min, and a ketone (1.0 mmol) was added to the resulting solution at 0 °C. The mixture was allowed to react for 12 h at this temperature, and then quenched with 1.0 mol·L⁻¹ NaOH. The aqueous phase was extracted with ethyl acetate (20.0 mL × 3), dried over MgSO_4 , filtered and concentrated. The residue was purified by column chromatography to give the tertiary alcohol. Enantiomeric excesses of products were determined by HPLC using suitable chiral columns from Daicel.

1-Phenyl-1-(thien-3-yl)ethanol (**3a**):^[7e] Yellow oil (179 mg, 88% yield). *ee* = 90% [Chiralpak OD-H column, $V(n\text{-hexane}) : V(i\text{-PrOH}) = 90 : 10$, flow rate = 1.0 mL/min, $\lambda = 254$ nm]. $t_{\text{R}}(\text{minor}) = 9.11$ min, $t_{\text{R}}(\text{major}) = 9.72$ min. ^1H NMR (300 MHz, CDCl_3) δ : 7.43~7.18 (m, 7H), 6.98 (s, 1H), 2.25 (s, H), 1.94 (s, 3H); ^{13}C NMR (75 MHz, CDCl_3) δ : 149.6, 147.4, 128.1, 127.0, 126.7, 126.0, 125.4, 120.7, 74.5, 31.1.

1-(4-Nitrophenyl)-1-(thien-3-yl)ethanol (**3b**): Yellow oil (227 mg, 91% yield). *ee* = 91% [Chiralpak AD-H column, $V(n\text{-hexane}) : V(i\text{-PrOH}) = 93 : 7$, flow rate = 1.0 mL/min, $\lambda = 254$ nm]. $t_{\text{R}}(\text{major}) = 35.537$ min; $t_{\text{R}}(\text{minor}) = 46.700$ min. ^1H NMR (300 MHz, CDCl_3) δ : 8.19~8.16 (m, 2H), 7.69~7.66 (m, 2H), 7.29~7.26 (m, 1H), 6.97~5.95 (m, 2H), 2.52 (s, 1H), 2.04 (s, 3H); ^{13}C NMR (75 MHz, CDCl_3) δ : 154.3, 151.7, 147.0, 126.8, 126.3, 125.8, 124.5, 123.4, 74.4, 32.1. HRMS (ESI) calcd for $\text{C}_{12}\text{H}_{11}\text{O}_3\text{NSNa}$ ($\text{M} + \text{Na}^+$) 272.0352, found 272.0349.

1-(2-Nitrophenyl)-1-(thien-3-yl)ethanol (**3c**): Yellow oil (224 mg, 90% yield). *ee* = 88% [Chiralpak OD-H column, $V(n\text{-hexane}) : V(i\text{-PrOH}) = 97 : 3$, flow rate = 1.0 mL/min, $\lambda = 254$ nm]. $t_{\text{R}}(\text{minor}) = 27.142$ min, $t_{\text{R}}(\text{major}) = 29.488$ min. ^1H NMR (300 MHz, CDCl_3) δ : 8.36~8.35 (m, 1H), 8.14~8.10 (m, 1H), 7.77~7.74 (m, 1H), 7.51~7.46 (m, 1H), 7.32~7.26 (m, 2H), 6.98~6.96 (m, 1H), 2.30 (s, H), 1.99 (s, 3H); ^{13}C NMR (75 MHz, CDCl_3) δ : 149.7, 148.3, 148.2, 131.7, 129.1, 126.9, 126.2, 122.0, 121.3, 120.4, 74.0, 31.2. HRMS (ESI) calcd for $\text{C}_{12}\text{H}_{11}\text{O}_3\text{NSNa}$ ($\text{M} + \text{Na}^+$) 272.0352, found 272.0351.

1-(Thien-3-yl)-1-(4-(trifluoromethyl)phenyl)ethanol (**3d**): Yellow oil (245 mg, 90% yield). *ee* = 90% [Chiralpak AD-H column, $V(n\text{-hexane}) : V(i\text{-PrOH}) = 93 : 7$, flow rate = 1.0 mL/min, $\lambda = 254$ nm]. $t_{\text{R}}(\text{major}) = 10.886$ min, $t_{\text{R}}(\text{minor}) = 14.053$ min. ^1H NMR (300 MHz, CDCl_3) δ : 7.62~7.57 (m, 4H), 7.27~7.26 (m, 1H), 6.97~6.93 (m, 2H), 2.49 (s, 1H), 2.02 (s, 3H); ^{13}C NMR (75 MHz, CDCl_3) δ : 152.3, 151.0, 126.7, 125.7, 125.4, 125.2, 125.1, 125.0, 124.4, 74.5, 32.1. HRMS (ESI) calcd for $\text{C}_{13}\text{H}_{10}\text{F}_3\text{S}$ ($\text{M} - \text{OH}^+$) 255.0450, found 255.0449.

1-(4-Bromophenyl)-1-(thien-3-yl)ethanol (**3e**): Yellow oil (232 mg, 82% yield). *ee* = 86% [Chiralpak OD-H col-

umn, $V(n\text{-hexane}) : V(i\text{-PrOH}) = 93 : 7$, flow rate = 1.0 mL/min, $\lambda = 254$ nm]. $t_{\text{R}}(\text{minor}) = 9.101$ min, $t_{\text{R}}(\text{major}) = 10.056$ min. ^1H NMR (300 MHz, CDCl_3) δ : 7.45~7.42 (m, 2H), 7.38~7.28 (m, 3H), 7.18 (s, 1H), 6.97~6.95 (m, 1H), 2.29 (s, H), 1.90 (s, 3H); ^{13}C NMR (75 MHz, CDCl_3) δ : 149.0, 146.4, 131.1, 127.2, 126.5, 126.2, 120.9, 74.1, 31.0. HRMS (ESI) calcd for $\text{C}_{12}\text{H}_{11}\text{BrS}$ ($\text{M} - \text{OH}^+$) 264.9681, found 264.9682.

1-(4-Chlorophenyl)-1-(thien-3-yl)ethanol (**3f**): Yellow oil (208 mg, 87% yield). *ee* = 90% [Chiralpak OD-H column, $V(n\text{-hexane}) : V(i\text{-PrOH}) = 97 : 3$, flow rate = 1.0 mL/min, $\lambda = 254$ nm]. $t_{\text{R}}(\text{minor}) = 15.427$ min, $t_{\text{R}}(\text{major}) = 18.464$ min. ^1H NMR (300 MHz, CDCl_3) δ : 7.44~7.40 (m, 2H), 7.31~7.23 (m, 3H), 6.94~6.89 (m, 2H), 2.42 (s, H), 1.99 (s, 3H); ^{13}C NMR (75 MHz, CDCl_3) δ : 152.8, 145.8, 133.0, 128.2, 126.8, 125.6, 125.2, 124.3, 74.4, 32.2. HRMS (ESI) calcd for $\text{C}_{12}\text{H}_{11}\text{ClSNa}$ ($\text{M} + \text{Na}^+$) 261.0111, found 261.0105.

1-(2-Chlorophenyl)-1-(thien-3-yl)ethanol (**3g**): Yellow oil (191 mg, 80% yield). *ee* = 90% [Chiralpak AD-H column, $V(n\text{-hexane}) : V(i\text{-PrOH}) = 93 : 7$, flow rate = 1.0 mL/min, $\lambda = 254$ nm]. $t_{\text{R}}(\text{major}) = 8.514$ min, $t_{\text{R}}(\text{minor}) = 9.488$ min. ^1H NMR (300 MHz, CDCl_3) δ : 7.81~7.78 (m, 1H), 7.33~7.23 (m, 4H), 6.91~6.90 (m, 1H), 6.76~6.74 (m, 1H), 2.33 (s, H), 2.09 (s, 3H); ^{13}C NMR (75 MHz, CDCl_3) δ : 152.4, 143.2, 131.3, 129.1, 127.6, 126.8, 126.6, 124.7, 124.1, 121.0, 74.5, 29.9. HRMS (ESI) calcd for $\text{C}_{12}\text{H}_{11}\text{ClSNa}$ ($\text{M} + \text{Na}^+$) 261.0111, found 261.0108.

1-(Thien-3-yl)-1-(*p*-tolyl)ethanol (**3h**): Yellow oil (175 mg, 80% yield). *ee* = 85% [Chiralpak OD-H column, $V(n\text{-hexane}) : V(i\text{-PrOH}) = 93 : 7$, flow rate = 1.0 mL/min, $\lambda = 254$ nm]. $t_{\text{R}}(\text{major}) = 14.415$ min, $t_{\text{R}}(\text{minor}) = 18.012$ min. ^1H NMR (300 MHz, CDCl_3) δ : 7.32~7.25 (m, 3H), 7.19~7.12 (m, 3H), 6.96~6.93 (m, 1H), 3.02 (s, 1H), 2.34 (s, 3H), 2.02 (s, 3H); ^{13}C NMR (75 MHz, CDCl_3) δ : 153.6, 144.3, 136.8, 128.7, 126.4, 125.2, 124.7, 124.0, 74.6, 32.1, 20.9. HRMS (ESI) calcd for $\text{C}_{13}\text{H}_{14}\text{OSNa}$ ($\text{M} + \text{Na}^+$) 241.0658, found 241.0658.

1-(Thien-3-yl)-1-(*o*-tolyl)ethanol (**3i**): Yellow oil (170 mg, 78% yield). *ee* = 85% [Chiralpak OD-H column, $V(n\text{-hexane}) : V(i\text{-PrOH}) = 93 : 7$, flow rate = 1.0 mL/min, $\lambda = 254$ nm]. $t_{\text{R}}(\text{major}) = 7.488$ min, $t_{\text{R}}(\text{minor}) = 8.388$ min. ^1H NMR (300 MHz, CDCl_3) δ : 7.73~7.69 (m, 1H), 7.25~7.21 (m, 3H), 7.15~7.12 (m, 1H), 6.91~6.88 (m, 1H), 6.76~6.74 (m, 1H), 2.30 (s, H), 2.14 (s, 3H), 2.03 (s, 3H); ^{13}C NMR (75 MHz, CDCl_3) δ : 153.2, 144.3, 136.4, 132.3, 127.8, 126.6, 125.5, 125.4, 124.6, 123.9, 74.8, 31.4, 21.3. HRMS (ESI) calcd for $\text{C}_{13}\text{H}_{14}\text{OSNa}$ ($\text{M} + \text{Na}^+$) 241.0658, found 241.0656.

1-(4-Methoxyphenyl)-1-(thien-3-yl)ethanol (**3j**): Yellow oil (194 mg, 83% yield). *ee* = 90% [Chiralpak OD-H column, $V(n\text{-hexane}) : V(i\text{-PrOH}) = 93 : 7$, flow rate = 1.0 mL/min, $\lambda = 254$ nm]. $t_{\text{R}}(\text{minor}) = 24.622$ min, $t_{\text{R}}(\text{major}) = 27.122$ min. ^1H NMR (300 MHz, CDCl_3) δ : 7.36~7.32 (m, 2H), 7.26~7.24 (m, 1H), 7.18~7.16 (m, 1H), 6.98~6.97 (m, 1H), 6.87~6.83 (m, 2H), 3.79 (s, 3H), 2.20 (s, H), 1.92 (s, 3H); ^{13}C NMR (75 MHz, CDCl_3) δ : 158.5, 150.0, 139.7,

126.6, 125.9, 120.5, 113.4, 74.3, 55.2, 31.1. HRMS (ESI) calcd for $C_{13}H_{14}O_2SNa$ ($M + Na^+$) 257.0607, found 257.0606.

1-(3-Methoxyphenyl)-1-(thien-3-yl)ethanol (**3k**): Yellow oil (199 mg, 85% yield). $ee=70\%$ [Chiralpak OD-H column, $V(n\text{-hexane}) : V(i\text{-PrOH}) = 93 : 7$, flow rate = 1.0 mL/min, $\lambda=254$ nm]. $t_R(\text{minor})=15.076$ min, $t_R(\text{major})=18.947$ min. 1H NMR (300 MHz, $CDCl_3$) δ : 7.25~7.22 (m, 2H), 7.08~7.02 (m, 2H), 6.92~6.89 (m, 2H), 6.82~6.79 (m, 1H), 3.79 (s, 3H), 2.49 (s, H), 2.00 (s, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ : 159.4, 153.2, 148.9, 129.1, 126.5, 124.9, 124.1, 117.8, 112.4, 111.4, 74.7, 55.2, 32.1. HRMS (ESI) calcd for $C_{13}H_{14}O_2SNa$ ($M + Na^+$) 257.0607, found 257.0611.

1-(Pyridin-3-yl)-1-(thien-3-yl)ethanol (**3l**): Yellow oil (162 mg, 79% yield). $ee=91\%$ [Chiralpak AS-H column, $V(n\text{-hexane}) : V(i\text{-PrOH}) = 90 : 10$, flow rate = 0.8 mL/min, $\lambda=254$ nm]. $t_R(\text{major})=15.840$ min, $t_R(\text{minor})=18.044$ min. 1H NMR (300 MHz, $CDCl_3$) δ : 8.63~8.62 (m, 1H), 8.45~8.43 (m, 1H), 7.78~7.74 (m, 1H), 7.30~7.20 (m, 3H), 6.99~6.97 (m, 1H), 1.96 (s, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ : 148.8, 148.1, 147.2, 142.9, 133.2, 126.5, 126.4, 122.9, 121.1, 73.2, 31.1. HRMS (ESI) calcd for $C_{11}H_{12}ONS$ ($M+H^+$) 206.0634, found 206.0629.

Supporting Information Copies of HPLC analytic data, and 1H NMR and ^{13}C NMR spectra of the compounds **3a**~**3l**. The Supporting Information is available free of charge via the Internet at <http://sioc-journal.cn/>.

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